

AI-POWERED CAD ANALYSIS REPORT

Generated: 12 Aug 2026

Geometry to Process to Cost · Fully AI-reasoned

100

Flange with Central Boss (6-Hole Bolt Circle)

80 × 80 × 20 mm · 63.9 cm³ · AI 0.173 kg / Steel 0.502 kg

Manufacturability: 100/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	80.0 × 80.0 × 20.0 mm	Surface Area	148.4 cm ²
Volume	63.94 cm ³	Weight (AI)	0.173 kg
Weight (Steel)	0.502 kg	Weight (Plastic)	0.067 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Bolt-circle holes	6	High	6x through-holes, r=3mm, on +X face
Central boss/shaft	1	High	Cylindrical boss, r=15mm, likely spigot or pilot register
Flange OD / register	1	Medium	Outer cylindrical face, r=40mm, defines flange envelope
Planar faces	3	Medium	3 flat faces forming flange front/back and boss shoulder

§3 — Material Analysis

AI-suggested

Primary Material:

Plain Carbon Steel (55% confidence)

No aluminium signal in filename, wall thickness (12-20mm), or geometry — thick-section flange with a central boss and bolt circle is typical of a structural/drivetrain-adjacent steel flange. Per guidance, defaulted to steel over aluminium when unsure. mat-dc01 selected as the closest generic machinable steel in the valid list; a forged 4140/EN8 billet would be the real-world equivalent.

Alternatives:

Stainless Steel 304 (25%) · Aluminium 6061 (15%) · Ductile Cast Iron (5%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
CNC Machining (VMC, 2 setups)	machining	90%	0.5820	User-forced machining commodity. Prismatic flange geometry (73% cylindrical faces, 27% planar) with a 2-setup approach (+X:8 features, +Z:3 features) is well suited to 3-axis milling with a 4th-axis or secondary flip for the boss face and bolt circle.

§5 — Manufacturability Risks (Score: 100/100)

Severity	Feature / Area	Description	Recommended Action
Low	Bolt circle holes	6 through-holes at r=3mm are straightforward drilled features with no tight tolerance flagged.	Use a single drill cycle with peck if depth-to-diameter exceeds 5:1.
Low	Boss OD (r=40mm) and shoulder	Zero-draft cylindrical faces (8 of 11 faces have zero draft) are normal for machined parts but confirm no as-cast/forged stock allowance is assumed.	Confirm stock form (bar vs. forging) to size roughing allowance correctly.

§6 — DFM Issues (machining)

Severity	Area	Description	Impact	Fix
Medium	Bolt circle (6x r=3mm holes)	Six small holes on a single face require tight positional tolerance for bolt-circle alignment; if not called out, risk of assembly mismatch.	Rework or scrap if positional tolerance not met; added CMM inspection time.	Specify true-position tolerance on drawing and use a drill jig/fixture with locating dowels to hold repeatability across 200k units.
Medium	Central boss (r=15mm) concentricity to flange OD (r=40mm)	Two setups (+X, +Z) with different reference cylinders risk concentricity error between boss and OD if datum scheme is not consistent.	Out-of-tolerance runout can cause functional issues if boss mates with a shaft or bearing.	Machine boss and OD in the same setup where possible, or use a common datum (e.g., one bore) to locate both operations.
Low	Wall thickness variation (11.95-19.95mm)	Non-uniform wall ($\bar{A} = 4$ mm) asymmetric stock removal, which can cause minor part distortion after roughing.	Slight dimensional drift, more likely on tighter-tolerance faces after heavy roughing cuts.	Rough symmetric passes first and finish critical faces last to minimize residual stress distortion.
Low	High-volume machining economics	At 200,000 units/year, 0.582hr/part cycle time (35min) is long for pure machining vs. cast+machine hybrid alternatives.	Machining-only cost at this volume will be dominated by machine-hour cost; may not be the cheapest process route despite being user-specified.	Consider cast-and-machine (e.g., ductile iron or aluminium casting with light machining) to cut cycle time and cost at this volume, if design allows.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC		MOST LIKELY		CONSERVATIVE	
£15.00		£22.50		£34.00	
Net Weight		0.502 kg		Material	
				steel	

Recommended Process	machining	Cycle Time	0.0824 hr/part
Setup Time	0.500 hr	Operations	3 ops
Operations Detail	Turning — +X (8 faces) (mach-lathe-cnc, 0.0396 hr) Turning — +Z (3 faces) (mach-lathe-cnc, 0.0148 hr) Drilling — 6 holes (6×Ø6.0×12) [geometry-measured] (mach-drill, 0.0280 hr)		

Process-Specific Parameters

Casting Subtype	sand
Die/Mould Cost	£0
Die Life	0 shots
Cavities	0
Yield	0.0%
Flash Weight	0.000 kg
Yield	0.0%
Die Cost	£0
Strokes	0
Cavities	0
Wall Thickness	0 mm
Mould Cost	£0
Mould Life	0 shots
Projected Area	0.0 cm²

§8 — AI Analysis Explanation

This is a compact flange (80x80x20mm) with a central boss and 6-hole bolt circle, machined from solid stock in 2 setups (+X for the bolt-circle face, +Z for the boss/OD face). Geometry is dominated by cylindrical faces (73%) consistent with a turned/bored boss and bored bolt holes, plus 3 planar faces for the flange faces. No aluminium signal was present (filename, wall thickness, or draft pattern), so per standing guidance for dense thick-section solids, steel was selected over aluminium as the default material assumption, with mat-dc01 used as the closest valid generic steel grade — a real-world equivalent would likely be a forged 4140/EN8 bar. All cycle time, setup count, weight, and manufacturability values were taken verbatim from the OCCT-derived bottom-up CNC estimate as instructed. Cost range reflects a labour+machine-hour dominated unit cost typical of a 2-setup VMC job at moderate-high annual volume.

§9 — Analysis Limitations & Assumptions

1. Material grade (mat-dc01) is inferred without any explicit metadata or drawing note — real material could be a stainless or cast iron flange grade.
2. Operations breakdown between the two setups is estimated proportionally; a real CAM program could shift time distribution between milling and drilling.
3. Cost range assumes generic UK machine-hour rates and does not include specific tooling wear-out costs for 200k-unit production, which may favor a cast-and-machine hybrid process not fully modeled here.

